

AI-Powered Adaptive 3D Escape Room

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1 Problem Definition

Current AI study applications commonly focus on a narrow workflow: students upload notes, then the system generates flashcards or short quizzes. While this improves content preparation efficiency, the learning interaction is often text-heavy, repetitive, and low in immersion, which can reduce sustained motivation over time.

From an educational perspective, these tools also present pedagogical limitations. First, question delivery is frequently uniform and does not sufficiently adapt to individual prior knowledge, causing either cognitive overload or under-challenge. Second, feedback is often limited to correct/incorrect labels, offering insufficient support for misconception repair, reasoning development, and long-term retention.

This project addresses these gaps by reframing the learning process as an AI-powered, question-driven 3D escape room. Instead of passively reviewing generated cards, students actively solve contextualized challenges embedded in interactive environments, promoting deeper cognitive engagement.

The proposed adaptive mechanism adjusts question type, topic, and difficulty according to learner performance while providing scaffolded hints and concise explanations. In addition, uploaded user materials (PDF notes, lecture slides, or textbook excerpts) are prioritized during question generation to align challenges with current coursework. Consequently, the system supports retrieval practice, conceptual reinforcement, and sustained participation, aligning game mechanics with measurable student learning outcomes.

2 Target User & Persona

2.1 Target Audience

The proposed system targets:

- University students (ages 18–25) who currently use AI study tools for note upload, flashcard generation, and auto-quiz practice
- Learners in content-heavy courses who need frequent retrieval practice (e.g., STEM, social science, and business subjects)
- Students who struggle to maintain motivation in text-only study workflows
- Users who respond well to interactive learning, short study sessions, and game-based progression

2.2 User Persona

Name: Alex Chen

Age: 20

Background: Second-year university student who studies with AI tools by uploading lecture notes to generate flashcards and quizzes.

Behaviors:

- Studies in short 15–30 minute sessions between classes
- Frequently uploads notes before quizzes to generate quick practice sets
- Completes generated flashcards rapidly, often without deep conceptual reflection
- Uses casual games to take breaks when studying becomes monotonous

Goals:

- Improve exam performance through consistent and effective daily practice
- Better understand concepts rather than memorizing isolated facts
- Receive immediate explanations that correct misunderstanding
- Stay motivated through interactive and rewarding study activities
- Build long-term study habits using short but meaningful learning sessions

Frustrations:

- Flashcards and basic auto-quizzes feel repetitive and disconnected from real problem solving
- Question difficulty is often poorly calibrated (too easy or too hard)
- Most tools provide limited feedback beyond correct/incorrect labels
- Motivation drops quickly in text-heavy study interfaces
- Weak topics are not always identified clearly across sessions

How the System Improves Alex's Learning:

- **Contextualized Practice:** Questions are embedded in 3D scenarios, turning passive review into active, problem-based learning.
- **Adaptive Sequencing:** The system adjusts domain, format, and difficulty using Alex's accuracy and hint dependency.
- **Explanatory Feedback:** Each response is followed by concise reasoning to support misconception repair and deeper understanding.
- **Mastery-Oriented Progress:** Topic-level performance tracking helps Alex identify and strengthen weak knowledge areas.

- **Sustained Engagement:** Collectible card sets, character unlock milestones, and future leaderboard competition promote consistent study participation.

The proposed system addresses these frustrations by transforming the conventional note-upload study workflow into an adaptive and immersive learning experience. For Alex, this means short study sessions become structured micro-learning cycles that improve both engagement and academic performance.

3 Proposed Features & Value Proposition

3.1 Core Functionalities

The proposed application is an AI-powered 3D escape room built as a web-based interactive experience. Key features include:

- **3D Multi-Theme Environment:** Users explore virtual scenes rendered with modern web-based 3D technologies. Initial themes include beach and classic escape room styles, with additional 3D models and theme packs planned.
- **Object-Based Interaction:** Users can click on objects (e.g., doors, safes, paintings) to trigger puzzles and clues.
- **Personalized Content Integration (User Materials):** Students can upload PDFs, lecture notes, or textbook excerpts to ground question generation in their own coursework.
- **Automated Material Analysis:** The system parses uploaded content into structured learning objectives and tags them by topic and difficulty.
- **AI-Generated Multi-Domain and Course-Aligned Questions:** Each session generates unique questions across multiple domains using a Large Language Model (LLM), while prioritizing uploaded materials when available.
- **Multiple Question Formats:** The AI can output different challenge types, including multiple-choice, short-answer, ordering, and clue interpretation questions.
- **Adaptive Hint System:** The system provides hints that adjust based on user progress and difficulty.
- **Learning Feedback Engine:** The system explains answer reasoning and highlights related concepts after each challenge.
- **Mastery Tracking:** Domain-level performance metrics are used to adapt future questions and target weaker knowledge areas.
- **Puzzle Completion Card Reveal:** Each completed puzzle grants one card reveal, encouraging students to finish challenges consistently.
- **Card Set Collection and Character Unlock:** Cards are organized into multiple themed sets. When a student completes an entire set, the corresponding character is unlocked automatically (without in-game currency).

- **Midjourney-Generated Card Art Pipeline:** Card visuals are generated using Midjourney, then curated and integrated as collectible set assets linked to character rewards.
- **Game Progression System:** A structured sequence of puzzles must be solved to unlock the final escape condition.

3.2 Planned Post-MVP Extensions

- **Account Login and Profiles:** Login will be added in a later phase to persist user progress, inventory, and statistics across sessions.
- **Leaderboard System:** Global and friend-based rankings will allow users to compete using completion time, score, and accuracy metrics.
- **Seasonal Content Expansion:** New themes, character sets, and challenge collections can be released as content updates.

3.3 Value Proposition

The system addresses the limitations of traditional escape room experiences by:

- Increasing replayability through dynamic question generation and rotating visual themes
- Bridging generic trivia and course-specific preparation through user-material-grounded challenges
- Enhancing engagement with adaptive difficulty and mixed question formats
- Providing immersive interaction through 3D exploration and object-based gameplay
- Encouraging long-term retention with collectible card-set progression and milestone-based character unlocks
- Supporting future social competition through leaderboard integration

3.4 Student Learning Impact

Beyond entertainment, the platform is designed as an interactive learning environment for university-age users. Learning is fostered through:

- **Retrieval Practice:** Frequent question-solving improves recall of key concepts.
- **Interleaved Learning:** Mixed domains and formats reduce rote memorization and improve transfer across topics.
- **Curriculum Alignment:** Questions grounded in uploaded course materials improve relevance and direct exam preparation value.
- **Scaffolded Support:** Adaptive hints provide guidance in steps, helping learners stay challenged but not overwhelmed.

- **Feedback for Reflection:** AI explanations help students understand misconceptions instead of only seeing correct/incorrect labels.
- **Long-Term Motivation:** Card-set collection milestones and social comparison features encourage repeated practice sessions.

For the persona Alex, this means short gaming sessions also become productive micro-learning sessions: he practices reasoning, receives immediate correction, and progressively improves weak areas while staying engaged.

4 Technical Architecture & AI Stack

4.1 Frontend

The frontend will be developed using:

- React (Vite) for application structure
- React Three Fiber (Three.js) for 3D rendering
- React-based HUD and modal components for questions, hints, inventory, customization, and material upload
- Client-side state management for theme selection, session progress, card collection display, and personalized question-bank status

4.2 Backend

The backend will be implemented using:

- Python FastAPI for API development
- Game state services for tracking puzzle progression and session outcomes
- Document ingestion services for extracting and chunking uploaded study materials
- Question orchestration logic for domain, type, and difficulty control
- Learning analytics services for tracking mastery by topic and hint dependency
- Card-reward and set-completion services for progression rewards
- Supabase integration for managed PostgreSQL storage and secure data access
- Future authentication and leaderboard modules for persistent profiles and competitive ranking

4.3 Data Layer

- Supabase (PostgreSQL) as the primary cloud database
- Storage for player profiles, scores, question history, card inventories, and set-completion unlock records
- Storage for uploaded material metadata, processed text chunks, and session-specific question-bank references
- Learning records including topic-level accuracy, response time, and hint usage history
- Row-level security policies to protect per-user progression and inventory data
- Optional use of Supabase Auth in the post-MVP login phase

4.4 AI Services

- **LLM API:**
 - Multi-domain and curriculum-grounded question generation with answer key creation
 - Hint generation with adaptive difficulty control
 - Explanation feedback to improve learning and clarity
 - Misconception-aware follow-up prompts to reinforce weak concepts
- **Prompt Engineering:**
 - Structured prompts to ensure consistent output format across question types
 - Domain-specific templates for different fields (STEM, humanities, and general knowledge)
 - Source-priority rules that favor uploaded user materials over generic knowledge when available
 - Controlled difficulty calibration based on player performance
 - Learning-objective tagging so generated questions align with target skills
- **RAG Pipeline for User Materials:**
 - Retrieval from uploaded notes and course documents to ground question generation
 - Retrieval of curated templates to improve factual quality and consistency
- **Midjourney-Assisted Image Generation:**
 - Generate collectible card-set illustrations and style references
 - Link each curated card set to a corresponding unlockable character
 - Human curation before integration into final game assets

4.5 System Flow

1. User starts a session (guest in MVP; login-enabled in a future phase)
2. User uploads optional study materials (e.g., a lecture PDF or textbook chapter)
3. Backend extracts text, segments content, and tags learning objectives by topic and difficulty
4. LLM and RAG services generate a personalized session question bank, prioritizing uploaded materials
5. User chooses a scene theme and enters the 3D environment
6. User interacts with a 3D object, triggering a challenge request
7. Backend selects an appropriate question from the personalized bank and calibrates difficulty
8. Challenge is displayed to the user through the frontend UI
9. User submits an answer; backend validates and updates progression state
10. Session progress, learning analytics, rewards, and unlock states are persisted in Supabase
11. The system provides explanation feedback and adaptive hints based on the learner's response
12. If correct, the system grants score and one card reveal; if incorrect, it increases support and retries
13. When a full card set is collected, the corresponding character is unlocked automatically; leaderboard updates are posted when account mode is enabled

5 Budget Estimation

5.1 Cost Breakdown

- HKUST API usage (development quota): \$0 HKD (expected during course period)
- Hosting (Vercel or equivalent free tier): \$0 for prototype
- Supabase (free tier for MVP): \$0 initially
- 3D assets (open-source + self-produced): \$0 to low one-time cost
- Midjourney subscription for concept generation: approximately US\$10–US\$30 per month
- Development tools: mostly free student/individual tiers

5.2 Justification

The main variable costs are LLM usage and Midjourney subscription, with Supabase remaining free in the MVP stage under typical usage limits. Costs can be controlled through output caching, generation throttling, and batch production of cosmetic assets. This keeps the architecture cost-efficient for a student project while still supporting future expansion.

6 Conclusion

This project proposes an AI-powered adaptive 3D escape room that combines personalized challenge generation, immersive themed environments, and long-term progression mechanics. By integrating modern web technologies with AI services, the system moves beyond static puzzle gameplay and delivers a more personalized and replayable user experience.

The MVP focuses on adaptive gameplay, card-set collection progression, and 3D immersion, while later phases introduce login and leaderboard competition. This phased approach is technically feasible, budget-conscious, and well aligned with user engagement goals.